Engineering Applications of Artificial Intelligence 136 (2024) 108993

Contents lists available at ScienceDirect

Engineering Applications of Artificial Intelligence

journal homepage: www.elsevier.com/locate/engappai

Research Paper

Generative adversarial networks enable outlier detection and property monitoring for additive manufacturing of complex structures

Alexander Henkes [a,c,*,1], Leon Herrmann [b,1], Henning Wessels [c], Stefan Kollmannsberger [d]

[a] Computational Mechanics Group, ETH Zürich, Switzerland
[b] Chair of Computational Modeling and Simulation, TU München, Germany
[c] Division Data-Driven Modeling of Mechanical Systems, TU Braunschweig, Germany
[d] Chair of Data Science in Civil Engineering, Bauhaus-Universität Weimar, Germany

ARTICLE INFO

Keywords:
Additive manufacturing
Property monitoring
Structural health monitoring
Computational homogenization
Deep learning
Neural networks

ABSTRACT

Although additive manufacturing processes have matured in many areas, difficulties with regard to printing accuracy persist. Possible defects are, *e.g.*, the generation of unwanted internal pores or a lack of fusion between layers. In general, defects result in a deviation between as-planned and as-built geometries. Assessing the influence of such deviations on effective (mechanical) properties requires either experimental testing or numerical investigation, both of which are often too complex for being used in time-critical production settings. Previous work with the Finite Cell Method (FCM) has shown that image-based computational homogenization can assist in principle in quality monitoring of produced parts by providing effective mechanical properties of as-built geometries. However, conducting FCM computations online for each and every part of a series production demands a prohibitive amount of computational resources.

The paper at hand suggests a remedy to this problem by using generative adversarial networks. This manuscript presents an integrated computational workflow for outlier detection and property monitoring, which in principle consists of four steps: (i) Carry out experiments under varying process conditions and collect respective (micro)-structural images. (ii) Compute effective properties for these images using FCM. (iii) Compose a generative adversarial network (GAN) training dataset from those images that meet desired effective properties. (iv) Use the GAN discriminator to detect outliers in series production. To this end, the discriminator is used as a classifier on as-built parts to judge whether an as-built structure is acceptable or defective. The viability of the approach is demonstrated on additively manufactured lattice structures whose geometry is acquired after production via computed tomography. The methodology is not only applicable for automated property monitoring but potentially also for reliability estimates of neural network-based property predictors.

1. Introduction

Additive manufacturing (AM) allows for the production of parts with complex topologies and, in particular, metamaterials. This is especially true for powder bed fusion (PBF) processes. A bottleneck is, however, the printing accuracy within the AM processes, in which inaccuracies can be caused by thermal distortions, lack of fusion, or precision limits. Thus, significant deviations between as-designed and as-built part geometries are typically present. The magnitude of such deviations is influenced by the tight interplay between the specific processing conditions (the process parameters) and the part geometries. Thus, both need to be planned carefully and individually before production. However, even careful planning cannot overcome the fact that AM production processes are inherently stochastic, such that as-designed parts will always differ from as-built parts. Ultimately, no part is free of defects. The key question is: Is the effect of such a defect relevant or not?

Answers can be provided by image-based simulation, see Evans et al. (2023) for a recent review. The usual steps in image-based simulation are:

(i) The acquisition of the as-built geometry of the part using scanning techniques such as laser scanning, structured light scanning, computed tomography (CT), Magnetic resonance imaging (MRI), or the like. This step is often followed by a segmentation procedure.

* Corresponding author at: Computational Mechanics Group, ETH Zürich, Switzerland.
E-mail address: ahenke@ethz.ch (A. Henkes).

[1] These authors contributed equally to this work and therefore share the first authorship.

https://doi.org/10.1016/j.engappai.2024.108993
Received 2 January 2024; Received in revised form 23 May 2024; Accepted 13 July 2024
Available online 30 July 2024
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(ii) The conversion of the image to a model suitable for analysis in computational mechanics, usually obtained by converting the domain of interest into lower resolutions and/or a mesh.
(iii) The computational mechanical analysis itself using a numerical method such as Finite Elements, Finite Volumes, Lattice Boltzmann, Finite Differences, or others. Given the large amount of data to be analyzed combined with often non-trivial physical behavior to be emulated, this step is particularly resource intensive. Additional resources are necessary for the subsequent data analysis and visualization.

Under the assumption that the effect of the defect does reflect in the overall (effective) properties of the produced part, computational homogenization is used to reduce the costs of a numerical evaluation of the mechanical quality of produced parts. The field of computational homogenization is vast, and we refer to Geers et al. (2017) for a basic as well as to March et al. (2023) for a more recent overview of methods used for computational homogenization to predict mechanical properties of additively manufactured parts. Some of these approaches have been specifically developed to reduce the computational cost involved in numerical homogenization. Among them are the Fast Fourier Transform-based homogenization (Schneider, 2021) or the Finite Cell Method (FCM) (Düster et al., 2012). Yet, depending on the production cost and use of the produced part, even these modern variants can lead to unfeasible computational costs.

In an attempt to further speed-up tasks involving repeated computational homogenization, homogenization approaches based on convolutional neural networks (CNN) have been developed, see, *e.g.*, Veres et al. (2015), Rao and Liu (2020), Henkes et al. (2021) and Mann and Kalidindi (2022) or Eidel (2023). Although physics-informed learning has already been considered in the context of micromechanics, see, *e.g.*, Henkes et al. (2022), to the best of the authors' knowledge, image-property maps in the context of computational homogenization, learned by CNNs are usually purely data-driven and thus physics-agnostic, although physics-informed variants exist in other contexts, *e.g.*, Zhao et al. (2023). Standard CNNs, thus, have difficulties in the extrapolation regime, *i.e.*, in predicting properties outside the dataset they have been trained on. Furthermore, in their original form, no mechanism exists to cheaply evaluate if a prediction is reliable or not or if the data under investigation stems from the original set the network was trained upon. In the present work, we refer to this task as outlier detection, which in the deep learning community is also known as anomaly or novelty detection.

Generally, outlier detection approaches rely on either basic statistical methods to detect out-of-distribution samples, *e.g.*, Euclidean or Mahalanobis distances (Lee et al., 2018), the Kullback–Leibler divergence, Gaussian mixture models (Reynolds, 2009), small perturbations to the input (Liang et al., 2020), or PCA-based reconstruction errors (Jolliffe and Cadima, 2016). Alternatively, models contain built-in extensions providing a probabilistic measure, such as classification probabilities acting as confidence scores (Hendrycks and Gimpel, 2018), ensemble methods (Lakshminarayanan et al., 2017; Mohammed and Kora, 2023), or Bayesian approaches (Gal and Ghahramani, 2016; Goan and Fookes, 2020; Jospin et al., 2022). The former methods are not necessarily tuned to the problem at hand, while the latter come at an additional computational cost or require knowledge of the outliers (see Berger et al., 2021 for a comparison of some of the methods).

For many anomaly detection tasks, deep learning-based approaches are now state-of-the-art, of which Pang et al. (2022) provides an overview of the prevalent methodologies. Therein, three main categories for deep learning-based anomaly detection are identified: (i) Deep learning for feature extraction, (ii) learning feature representations of normality, and (iii) end-to-end anomaly score learning.

In the first, conventional non-linear dimensionality reduction techniques, such as autoencoders (Hinton and Salakhutdinov, 2006) are employed, such that simpler anomaly measures (Hodge and Austin, 2004; Chandola et al., 2009; Aggarwal, 2016; Boukerche et al., 2020), *e.g.*, clustering or support vector machines, can be employed on the identified low-dimensional space (Erfani et al., 2016; Ionescu et al., 2017; Xu et al., 2017; Yu et al., 2018; Ionescu et al., 2019).

Learning feature representations of normality quantifies the anomalies indirectly. First, a feature representation is learned with a neural network, which subsequently is evaluated to determine an anomaly score. A straightforward implementation is, for example, enabled by autoencoders, where the data is first reduced to a small latent space and from there reconstructed. After training the autoencoder on normal data, *i.e.*, without anomalies, it can be applied to previously unseen anomalous data, which leads to larger reconstruction errors, *i.e.*, differences between input and output of the autoencoder (Hawkins et al., 2002; Zhou and Paffenroth, 2017; Lu et al., 2017; Xu et al., 2017; Kieu et al., 2019). This can similarly be achieved by generative adversarial networks (GANs) (Schlegl et al., 2017; Xia et al., 2022), where after training with normal data, the difference between the generator output and one sample is minimized with respect to the generator input. If it is not possible to minimize the difference sufficiently, the sample must be an anomaly. To avoid the additional optimization during the anomaly score evaluation, the inverse mapping of the generator can be learned during training of the GAN (Zenati et al., 2018; Schlegl et al., 2019), for example, using bi-directional GANs (BiGANs) (Donahue et al., 2017). Alternatively, the latter two concepts can be combined using variational autoencoder GANs (Akcay et al., 2019), exploiting the superior approximation power of GANs while avoiding the optimization with respect to the input during evaluation. Further variations rely on unsupervised classification of the normal data, where an anomaly is detected if a sample cannot be classified (Golan and El-Yaniv, 2018; Wang et al., 2019).

Lastly, also end-to-end anomaly scores can be established by training neural networks to directly predict anomaly scores. The most prominent method relying on generative approaches is to use the discriminator output as anomaly score (Sabokrou et al., 2018; Zheng et al., 2019; Ngo et al., 2019). After a GAN has been trained on normal (healthy) data, the discriminator is able to detect distinguish between normal and anomalous data. This is particularly practical if the generator is simultaneously needed for data generation of normal data, as, *e.g.*, when training a surrogate model. Then, the discriminator is essentially a byproduct of the GAN training. Therefore, we will focus on the last approach and investigate the viability of using the discriminator of a Wasserstein GAN for outlier detection and property monitoring for additively manufactured structures.

To this end, we propose an integrated computational pipeline, which ultimately allows to employ knowledge from computational methods in a straightforward manner not only in the design phase of the product life cycle but also to inform process monitoring during operation. The main steps of the presented framework are as follows:

(i) Using topology optimization (Bendsøe and Sigmund, 2004) and its variants, complex structures and metamaterials are computationally designed. In this work, such schemes are not considered, and we restrict ourselves to lattice structures.
(ii) Suitable process parameters are identified by experiments and the assessment of effective mechanical properties using image-based simulation, here FCM.
(iii) For series production, process instabilities necessitate a reliable monitoring. In this context, CNN homogenization has been proposed as a cost-effective surrogate for image-based simulation. To augment the experimentally available imaging data for CNN training, artificial images are needed, which can be obtained from GAN generators, see, *e.g.*, Henkes and Wessels (2022a).
(iv) To close the gap of CNN reliability indication and property monitoring in the context of AM, we propose to use the discriminator of the same GAN that has been used in (iii) for data augmentation. Unlike all previously mentioned approaches this does not cause additional computational effort within the proposed computational pipeline.

Note that although we introduce the Wasserstein GAN discriminator as a possible metric for CNN quality control, we do not explicitly consider CNNs in the present study. To investigate the relationship between Wasserstein GAN discriminator output and effective properties, we verify our approach by directly using numerical results.

The remainder of this article is structured as follows: First, we introduce the experimental, numerical and machine learning methods employed, namely micro CT (μCT) imaging (Section 2.1), the FCM (Section 2.2), the computational homogenization framework (Section 2.3), GANs (Section 2.4), and lastly GANs for anomaly detection (Section 2.5). Numerical results for an experimental test case and manufactured examples are presented in Section 3. The paper concludes in Section 4.

2. Methods

The interdisciplinary approach proposed in this manuscript combines experimental microstructural imaging, numerical analysis, and machine learning. Details on the experimental CT scanning setting employed are provided in Section 2.1. The obtained microstructural images are characterized by computing their effective mechanical properties using computational homogenization with the FCM. Details can be found in Sections 2.2 and 2.3 and references therein. The architecture of the GAN is outlined in Section 2.4. Finally, after the presentation of the individual methods, their interplay is described in Section 2.5.

2.1. CT scanning

CT scanning (Cormack, 1963; Hounsfield, 1973) is a useful tool in identifying the differences between as-designed geometries, *i.e.*, the original CAD geometry, and as-built geometries. The as-built geometries are obtained via CT scanning, a method relying on X-rays from various angles to map the object's internal structure. The X-ray beams are absorbed to different extents based on the varying densities within the object, creating a mapped density profile stored as voxel data. Specifically, the scans, reused from Korshunova et al. (2021b), were obtained with a Phoenix V CT scanner with a voxel resolution of 27 μm. A basic thresholding to determine the voxels with and without material has been performed. For more advanced segmentation procedures for CT scanned metal AM structures, see Korshunova et al. (2020).[2]

2.2. Finite cell method

The FCM (Parvizian et al., 2007; Düster et al., 2017), an immersed boundary method, embeds the structure of interest (defined in terms of a physical domain Ω) in an embedded domain Ω_e (see Fig. 1). The embedded domain Ω_e has a simple shape, such as a rectangle. Subsequently, this embedded domain Ω_e is meshed with a structured Cartesian grid. The task of representing the geometry is thus shifted from discretization to the numerical integration of the weak form of the boundary value problem. This is accomplished by introducing an indicator function $\alpha(\boldsymbol{x})$, defined as

$$\alpha(\boldsymbol{x}) = \begin{cases} 1, \forall \boldsymbol{x} \in \Omega, \\ \epsilon, \forall \boldsymbol{x} \in \Omega_e \setminus \Omega. \end{cases} \tag{1}$$

Here, $0 < \epsilon \ll 1$ is a small but non-zero numerical parameter to avoid ill-conditioning. The geometrical information is incorporated in the discretization through a multiplication of the indicator function $\alpha(\boldsymbol{x})$ with the weak form. To capture the geometry accurately, the integration points are distributed within each voxel. Due to the regular structure of the voxels and cells, the integrals can efficiently be pre-computed (Yang et al., 2012). The ansatz space of the cells can be chosen freely, where this work employs integrated Legendre polynomials of degree $p = 2$. Examples of FCM applied to CT scans in the literature range from AM (Korshunova et al., 2020, 2021b,a,c), biomechanics (Elhaddad et al., 2018; Hug et al., 2022a) to fracture in geology (Hug et al., 2022b). While the FCM is particularly advantageous if large structures or very high resolutions are considered, we do not harvest this feature in the context of this contribution. Instead, we use FCM for homogenization as presented in Korshunova et al. (2020) and rely on its feature to be able to accurately homogenize domains whose boundaries contain voids.

2.3. Computational homogenization

The goal of numerical homogenization in computational mechanics is to estimate the effective material tensor $\bar{\boldsymbol{C}}$ of representative volume elements (RVEs) within larger structures. To this end, the microscopic stresses $\boldsymbol{\sigma}$ and strains $\boldsymbol{\varepsilon}$ of the RVE can be related to macroscopic, *i.e.*, effective stresses $\bar{\boldsymbol{\sigma}}$ and strains $\bar{\boldsymbol{\varepsilon}}$. This is accomplished with the Hill–Mandel principle of strain energy density equality (Nemat-Nasser and Hori, 1993; Zohdi and Wriggers, 2008; Gross and Seelig, 2011), where a volume integral over the RVE domain Ω is performed

$$\frac{1}{2}(\bar{\boldsymbol{\sigma}} : \bar{\boldsymbol{\varepsilon}}) = \int_{\Omega} (\boldsymbol{\sigma} : \boldsymbol{\varepsilon}) d\Omega. \tag{2}$$

The volume integral can be transformed into a boundary integral over the RVE boundary Γ, yielding

$$\frac{1}{\Omega} \int_{d\Gamma} (\boldsymbol{u} - \langle \boldsymbol{\varepsilon} \rangle_{\Omega} \boldsymbol{x}) \boldsymbol{\sigma} - \langle \boldsymbol{\sigma} \rangle_{\Omega} \boldsymbol{n} \, d\Gamma = 0, \tag{3}$$

where $\langle \cdot \rangle_{\Omega}$ denotes the averaging operator over the domain Ω, and $\boldsymbol{n}$ is the normal vector of the boundary Γ. The condition from Eq. (3) can be satisfied a priori through specific boundary conditions, such as kinematic uniform boundary conditions, static uniform boundary conditions, or periodic boundary conditions (Pahr and Zysset, 2008). In this work, periodic displacement boundary conditions (Tian et al., 2019) are employed

$$\boldsymbol{u}(\boldsymbol{x}^+) - \boldsymbol{u}(\boldsymbol{x}^-) = \bar{\boldsymbol{\varepsilon}} \Delta \boldsymbol{x}, \tag{4}$$

where $\boldsymbol{x}^+$ and $\boldsymbol{x}^-$ represent coupled coordinates of opposing faces of the RVE, $\Delta \boldsymbol{x}$ is the distance between the two faces, and $\boldsymbol{u}$ is the displacement. To ensure the coupling of the displacement degrees of freedom at opposing faces, a periodic mesh must be employed. The Cartesian grid used within the FCM automatically fulfills this periodicity requirement.

After performing a numerical simulation on the RVE with periodic boundary conditions, the fluctuating microscopic stresses $\boldsymbol{\sigma}$ and strains $\boldsymbol{\varepsilon}$ are related to the corresponding macroscopic quantities $\bar{\boldsymbol{\sigma}}$, $\bar{\boldsymbol{\varepsilon}}$, which on average have to be equal. This is referred to as the micro-to-macro transition and can be written as

$$\bar{\boldsymbol{\sigma}} = \int_{\Omega} \boldsymbol{\sigma} d\Omega = \frac{1}{\Omega} \int_{\Gamma} \boldsymbol{t} \otimes \boldsymbol{x} d\Gamma, \tag{5}$$

$$\bar{\boldsymbol{\varepsilon}} = \int_{\Omega} \boldsymbol{\varepsilon} d\Omega = \frac{1}{\Omega} \int_{\Gamma} \frac{1}{2} (\boldsymbol{u} \otimes \boldsymbol{n} + \boldsymbol{n} \otimes \boldsymbol{u}) d\Gamma, \tag{6}$$

where $\otimes$ is the Kronecker product. The boundary integrals across voids pose difficulties for conventional numerical approaches. With the FCM, the voids in the CT scan data are treated as extremely soft materials using the indicator function $\alpha(\boldsymbol{x})$ from Eq. (1). As a consequence, Eqs. (5) and (6) remain valid, and the integration can directly be performed over the entire boundary $\partial\Omega_e = \Gamma_e$, including those parts of the boundary containing voids. The notation then becomes involved, and we refer to Korshunova (2021) for an in-depth presentation. The macroscopic stresses $\bar{\boldsymbol{\sigma}}$ and strains $\bar{\boldsymbol{\varepsilon}}$ can finally be related through the effective material tensor $\bar{\boldsymbol{C}}$ with

$$\bar{\boldsymbol{\sigma}} = \bar{\boldsymbol{C}} \bar{\boldsymbol{\varepsilon}}. \tag{7}$$

[2] We note that obtaining CT scans in a classical manner by scanning an artifact after it was produced might be too costly on an industrial scale. However, cost-effective alternatives exist, see Feng et al. (2022), which use data obtained from monitoring the production process to construct CT scans on the fly.

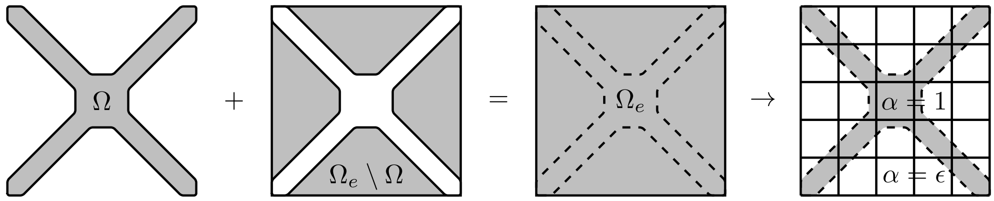


Fig. 1. Conceptual illustration of the FCM showing embedding and discretization with a Cartesian grid.
Source: Adapted from Parvizian et al. (2007).

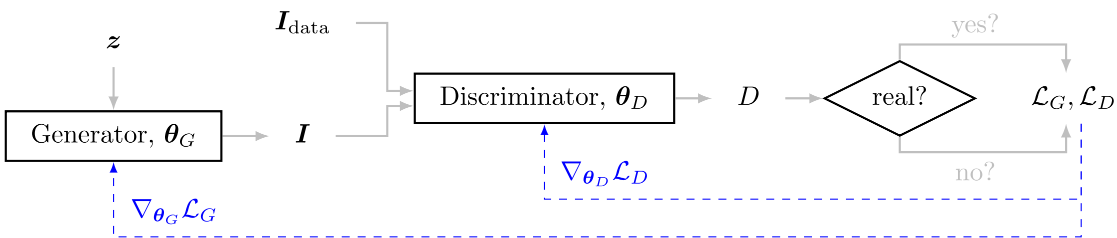


Fig. 2. Schematic concept of a Generative Adversarial Neural Network. A generator network is trained to generate as-realistic-as-possible artificial as-built geometries from a random vector z. The discriminator network is trained to distinguish between real I_{data} and fake as-built geometries I. The training is performed by minimizing the loss functions $\mathcal{L}_G, \mathcal{L}_D$, using the correct or incorrect classifications of the discriminator, *i.e.*, the scalar discriminator output D.

By applying six linearly independent load cases to the RVE, the effective material tensor $\bar{C}$ can be computed from Eq. (7). For further details of the employed first-order homogenization, see Korshunova et al. (2020).

2.4. Wasserstein GAN

Originally, GANs have been developed to create synthetic images $I \in \mathcal{I}$ which have the same statistical properties as real images $I_{\text{data}} \in \mathcal{I}_{\text{data}}$. In this work, the GAN's aim is to create synthetic three-dimensional images in the form of as-built AM geometries I. The training data set $\mathbb{D}$ is a subset of the real as-built geometry image space $\mathcal{I}_{\text{data}}$ and is defined by

$$\mathbb{D} = \{I^k_{\text{data}},\ k = 1, \ldots, n_s\}, \tag{8}$$

where n_s is the number of as-built geometries images inside the data set. A GAN consists of two competing neural networks $\mathcal{N}$, namely a generator G and a discriminator D. While the parameters θ_G of the generator network are optimized to maximize the discriminator's score on generated as-built geometries (with the goal of generating synthetic as-built geometries as close to the data set as possible), the discriminator is trained to distinguish between the latter by adjusting its own parameters θ_D. The overall architecture is sketched generically in Fig. 2.

Algorithm 1 GAN training

Require: data set $\mathbb{D}$ from Eq. (8)
Require: Generator G from Eq. (9)
Require: Discriminator D from Eq. (10)
  for i in n_{iter} **do**
    Sample random input vector z (see Eq. (9))
    Generate as-built geometry $G(z) = I$
    Sample real as-built geometry I_{data} from data set $\mathbb{D}$
    Calculate generator loss $\mathcal{L}_G$ from Eq. (16)
    Calculate discriminator loss $\mathcal{L}_D$ from Eq. (13)
    Update weights θ_G and θ_D as in Eq. (17)
  end for

The input to the generator $G = \mathcal{N}_G(z;\theta_G)$ is a random vector z. Hence, the generator maps random input vectors z from a d_Z-dimensional input space $\mathcal{Z} \subset \mathbb{R}^{d_Z}$ to as-built geometries I in the d_I-dimensional output space $\mathcal{I} \subset \mathbb{R}^{d_I}$, such that

$$G : \mathcal{Z} \subset \mathbb{R}^{d_Z} \to \mathcal{I} \subset \mathbb{R}^{d_I},\quad z \mapsto G(z) = I. \tag{9}$$

In turn, the discriminator $D = \mathcal{N}_D(I;\theta_D)$ maps a three-dimensional as-built geometry I to a scalar, such that

$$D : \mathcal{I} \subset \mathbb{R}^{d_I} \to \mathbb{R},\quad I \mapsto D(I). \tag{10}$$

The scalar output of the discriminator D expresses a measure of similarity between an input as-built geometry $I \in \mathcal{I}$ and original as-built geometries $I_{\text{data}} \in \mathbb{D} \subset \mathcal{I}_{\text{data}}$ contained in the data set $\mathbb{D}$ Eq. (8). This similarity measure is constructed by approximating a distance function, which in the present work is the Wasserstein distance W, defined as

$$W : \mathcal{I} \times \mathcal{I}_{\text{data}} \to \mathbb{R},\quad (I, I_{\text{data}}) \mapsto W(I, I_{\text{data}}) = \inf \mathbb{E}\left[\|I - I_{\text{data}}\|_1\right]. \tag{11}$$

Herein, $\mathbb{E}$ denotes the expectation operator. Since Eq. (11) is not computationally feasible, its approximation according to the Kantorovich–Rubinstein theorem, see Villani (2009), is employed

$$W(I, I_{\text{data}}) \approx \tilde{W}(I, I_{\text{data}}) = \sup_{\|D\|_L \le 1} \left(\mathbb{E}_{I\sim\mathcal{I}}[D(I)] - \mathbb{E}_{I_{\text{data}}\sim\mathcal{I}_{\text{data}}}\left[D(I_{\text{data}})\right] \right), \tag{12}$$

where $\|D\|_L \le 1$ denotes the Lipschitz-1 continuity of the discriminator. The distance $\tilde{W}(I, I_{\text{data}})$ Eq. (12) is then used to formulate both the loss functions of the generator and the discriminator, respectively.

The **discriminator loss** $\mathcal{L}_D$ consists of two terms, the first representing the approximated Wasserstein distance $\mathcal{L}_{\tilde{W}}$ Eq. (12) and the second a gradient penalty term $\mathcal{L}_{\text{GP}}$

$$\mathcal{L}_D = \mathcal{L}_{\tilde{W}} + \mathcal{L}_{\text{GP}}, \tag{13}$$

with

$$\mathcal{L}_{\tilde{W}} = \mathbb{E}_{I\sim\mathcal{I}}[D(I)] - \mathbb{E}_{I_{\text{data}}\sim\mathcal{I}_{\text{data}}}\left[D(I_{\text{data}})\right], \tag{14}$$

$$\mathcal{L}_{GP} = \lambda \mathbb{E}_{\hat{I}\sim\hat{\mathcal{I}}}\left[\left(\|\nabla_{\hat{x}} D(\hat{I})\|_2 - 1\right)^2\right]. \tag{15}$$

A sample from $\hat{\mathcal{I}}$ is denoted as $\hat{I}$, which is a uniform distribution interpolated between $\mathcal{I}_{\text{data}}$ Eq. (8) and $\mathcal{I}$ Eq. (9), while λ is a weighting term. Optimizing loss functions Eq. (14) and Eq. (16) will result in a generator, which produces images (in this case, three dimensional microstructures) minimizing the Wasserstein distance. The trained discriminator (or critic) on the other hand, approximates the real Wasserstein distance of the generated images and the dataset. Rigorous proofs of these characteristics for the generator and discriminator can be found in Arjovsky et al. (2017). Note that the gradient penalty term $\mathcal{L}_{GP}$ ensures Lipschitz-1 continuity of the discriminator, as described in Gulrajani et al. (2017), and is necessary for the dual formulation of the Wasserstein distance to be valid. A proof can be found in Villani (2009). This technique is known as *Wasserstein-GAN with Gradient Penalty* or *WGAN-GP* (Gulrajani et al., 2017). In the following, the abbreviation GAN describes a WGAN-GP.

The **generator loss** $\mathcal{L}_G$ is identical to the negative version of the first term in Eq. (14) and defined as:

$$\mathcal{L}_G = -\mathop{\mathbb{E}}_{\tilde{I}\sim\tilde{\mathcal{I}}}[D(\tilde{I})]. \tag{16}$$

During training, the generator is updated in the distance-minimizing gradient direction, whereas the discriminator is updated in the distance-maximizing gradient direction, *i.e.*,

$$\min_{\theta_G} \mathcal{L}_G, \qquad \min_{\theta_D} \mathcal{L}_D. \tag{17}$$

The optimization is typically carried out by mini-batch gradient descent with a learning rate α and a batch size n_B, such that the mean operators in the loss functions Eq. (14) and Eq. (16) act over the respective batches. The training algorithm is summarized in Algorithm 1. Details with respect to the training, including training stability and the choice of hyperparameters, can be found in Henkes and Wessels (2022a) and references therein. The hyperparameters of the network used in the present manuscript are specified in Table 1 and Section 3.

It has been reported earlier in Henkes and Wessels (2022a) that the employed GAN architecture is able to generate microstructures that mimic realistic samples of the training data set $\mathbb{D}$. While the generated microstructures could be employed further for CNN homogenization, see, *e.g.*, Henkes et al. (2021), the numerical results reported within this contribution aim at illustrating the discriminator's ability to detect geometric outliers.

2.5. GANs for anomaly detection of as-built structures

Remark: Section 2.5 is completely new. To enhance readability, we refrain from coloring changes in this section.

Our aim is to train a discriminator as defined in Eq. (10) that detects outliers. To this end, outliers are defined as parts that do not meet desired effective properties. Fig. 3 provides a technical flowchart of the proposed methodology, which in principle consists of 4 steps:

(i) **Experiment:** Carry out experiments under varying process conditions. Note that such experiments are often necessary anyways to identify optimal process parameter. Record images of the as-built geometries.
(ii) **Simulation:** Compute the effective properties for these images. In this contribution, we used the FCM and focused on mechanical properties. Note that any other effective properties, obtained with any other computational method can equally be taken into account.
(iii) **Training:** Compose a GAN training dataset $\mathbb{D}_{int}$ from those images that meet desired effective properties.
(iv) **Prediction:** Use the GAN discriminator to detect outliers in series production, bypassing numerical simulations in the online phase.

As a baseline to assess the accuracy of the GAN discriminator, we consider the results obtained via FCM. While using FCM directly is certainly more reliable than the proposed discriminator, the related computational cost makes it unfeasible to analyze each and every part in a series production.

Table 1
Hyperparameters used in all experiments, if not indicated otherwise, for the generator G from Eq. (9) and the discriminator D from Eq. (10).

Hyperparameter	Generator	Discriminator
Batch size n_B	8	8
Learning rate α	5×10^{-5}	1×10^{-4}
Nadam moment β_1	0.9	0.9
Nadam moment β_2	0.999	0.999
LeakyReLU γ	0.2	0.2
Gradient penalty factor λ	–	10
No. layers n_L	5	5
Kernel size n_k	3	3
Initialization of $\boldsymbol{\theta}$	orthogonal	orthogonal
Input random vector $\boldsymbol{z}$	$\mathcal{N}(\mathbf{0},\mathbf{1})$	–
Dim. random vector d_Z	128	–
Dim. mapping network d_w	128	–
No. layers mapping network n_{Lw}	8	–
No. filters n_f	64	8

Certainly, the generation of training data and the GAN training itself constitute a computational overhead, which needs to be justified. For individual products and small series, the presented methodology is therefore not beneficial. However, for large-scale series production of components for safety-critical industrial applications, such as aerospace, monitoring and certification remain a critical bottleneck for the implementation of additively manufactured parts. Here, the proposed workflow and methodology are expected to yield a significant contribution.

3. Numerical experiments

Remark: The entire Section 3 has been rearranged and edited. To enhance readability, we refrain from coloring changes in this section.

In the present manuscript, two different types of microstructures are considered and depicted in Fig. 4. Fig. 4(a) shows the first microstructure, which is a real-world metallic lattice microstructure designed in CAD and subsequently additively manufactured using laser-based PBF. The as-designed CAD model deviates in various aspects from the as-built structure. The effects of these deviations on effective properties (*i.e.*, the homogenized Young's moduli $\bar{E}_1, \bar{E}_2, \bar{E}_3$ in each spatial dimension obtained from $\bar{\boldsymbol{C}}$) have been studied earlier in Korshunova et al. (2021b,a) who employed the FCM for the same type of microstructure. The FCM will thus be considered as baseline method to assess the validity of our GAN-based approach. As will be demonstrated in Section 3.1, the proposed methodology yields excellent results in this real-world experiment.

In the subsequent Section 3.2, we thoroughly test the limits and potential pitfalls of this approach. For this purpose, we design and analyze specific artificial microstructures: First, localized artificial defects that mimic a cut are introduced into the real-world lattice structures. Second, a fully artificial microstructure, namely a matrix with spherical inclusions as depicted in Fig. 4(b) is considered.

In both test cases, the material is considered to be linear elastic, with Young's modulus and Poisson's ratio as $(E, \nu) = (1\,000$ GPa, $0.4)$. It must be mentioned that the proposed methodology is not restricted to linear elasticity only, as the numerical computations are only needed to classify as-built geometries as intact or defective. Moreover, the classification can also consider other than mechanical effective properties, *e.g.*, electric, magnetic or chemical properties. The data sets used in the following investigations are publicly available, as stated in the data availability section. Furthermore, the hyperparameters employed during training are listed in Table 1.

3.1. Additively manufactured lattice structure

We begin by studying a real microstructure with realistic manufacturing defects. Specifically, an as-built lattice structure, see Fig. 5,

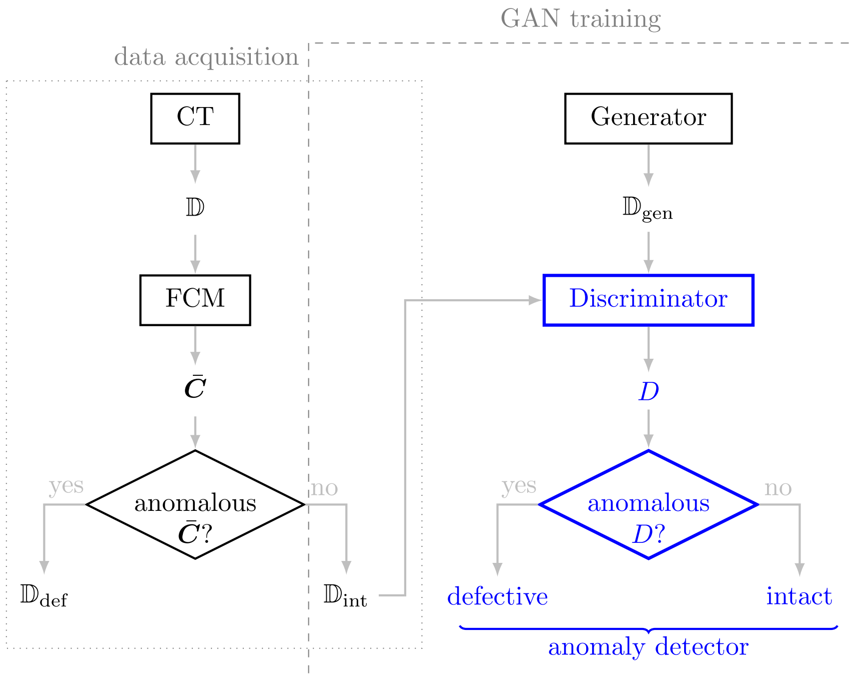


Fig. 3. Overview of the proposed anomaly detector (in blue) for as-built AM structures. As-built geometries are acquired through CT scanning (Section 2.1) and collected in the data set $\mathbb{D}$. Subsequently, each structure is computationally homogenized (Section 2.3) using FCM (Section 2.2) in order to obtain effective properties $\bar{\boldsymbol{C}}$ from Eq. (7). Upon the effective properties, the data set $\mathbb{D}$ is split into two subsets of intact samples $\mathbb{D}_{\text{int}} \subset \mathbb{D}$ and defected samples $\mathbb{D}_{\text{def}} \subset \mathbb{D}$. Only the intact samples are used for GAN training (Section 2.4). After training, the GAN discriminator is used to classify structures as intact or defective by providing an anomaly score. For this purpose, a threshold D^c needs to be defined.

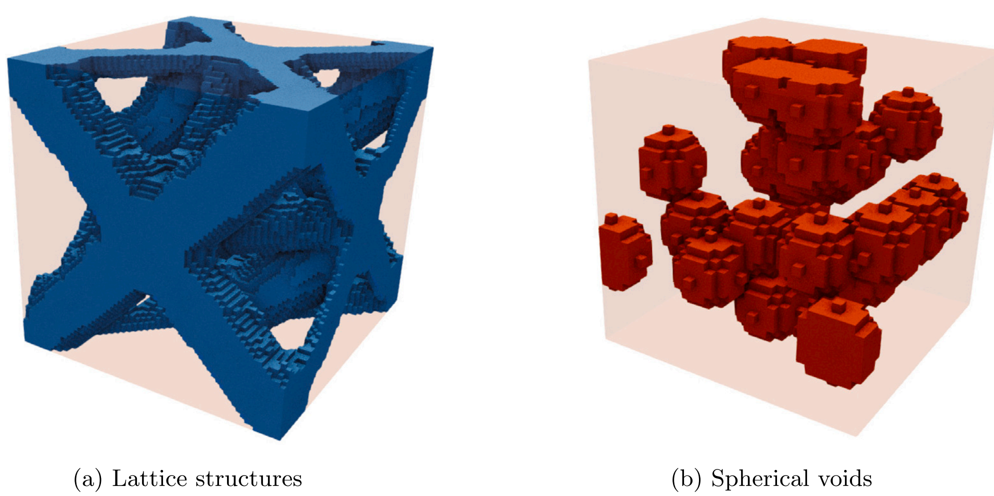


Fig. 4. Microstructures studied in this manuscript. First, the proposed methodology is assessed on real-world lattice structures (blue represents material) in Section 3.1. Artificial microstructures with spherical voids (red represents the voids) are examined in Section 3.2.2 to showcase potential limitations and pitfalls.

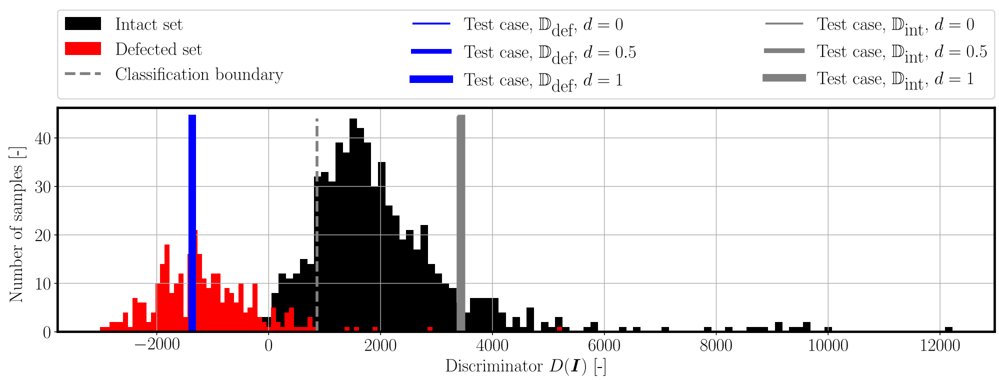
obtained from a real CT scan (Korshunova et al., 2021b,a) is considered. As reported in Korshunova et al. (2021b), the specimen was produced with laser-based PBF using SS316L metal powder on a Renishaw AM400 PBF system in the laboratory 3DMetal@UniPV. The detailed process parameters are provided in Table 2.

According to the flowchart in Fig. 3, the first step is to compose a training data set $\mathbb{D}_{int}$ of as-built geometries classified as intact. For this purpose, 1 000 randomly selected substructures $\mathbb{D}$ with $64 \times 64 \times 64$ voxels (corresponding to the lattice unit cell size) are extracted, see Fig. 4(a) for an example substructure. Subsequently, the substructures are treated as RVEs and are homogenized with the homogenization framework described in Section 2.3. The distribution

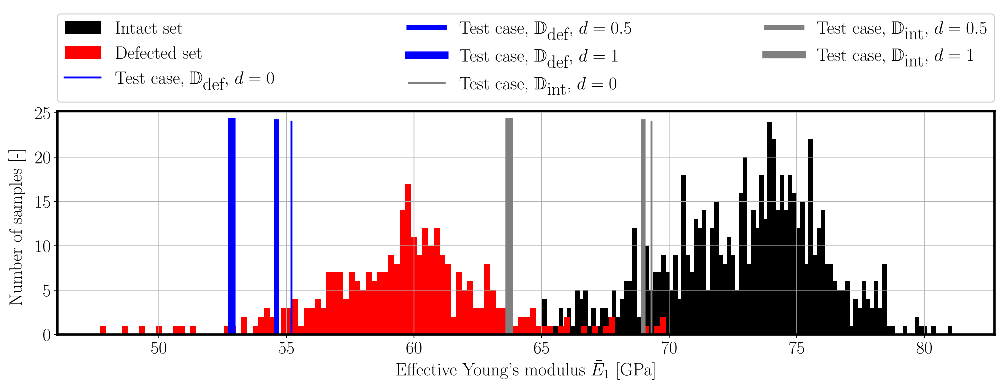
Table 2
Process parameters for additively manufactured lattice structure depicted in Fig. 5. Adapted from (Korshunova et al., 2021b).

Process parameters	Value
Build plate temperature	170 °C ± 1 °C
Chamber temperature	35 °C ± 5 °C
Layer thickness	50 μm ± 1 μm
Hatch spacing	110 μm ± 2 μm
Scan speed	1200 mm/s ± 2 mm/s
Laser power	200 W ± 0.1 W
Laser spot size	70 μm ± 1 μm

Fig. 5. As-built lattice structure created with metal PBF.



(a) Discriminator evaluation



(b) Effective Young's moduli

Fig. 6. Comparison of the distribution of discriminator evaluations (a) and effective Young's moduli in the x_1 direction (b). The defected and intact set consist of substructures from a CT-scanned lattice structure. The structures with effective properties below pre-defined thresholds are assigned to the defected set, while the others are deemed sufficiently good and thereby considered as intact. In addition, two manufactured test cases with a disconnected lattice beam $d = [0, 1]$ are considered. One manipulated structure is selected from the defected set $\mathbb{D}_{\text{def}}$, while the other originates from the intact set $\mathbb{D}_{\text{int}}$.

of the effective Young's moduli in x_1-direction is given in Fig. 6(b). Only structures with effective properties within desired bounds are to be considered during the GAN training. Therefore, substructures with effective properties lower than a certain threshold are considered defective. In this work, the defective set $\mathbb{D}_{\text{def}}$ is found with the following thresholds:

$$\bar{E}_1 < 65 \text{ GPa}, \tag{18}$$

$$\bar{E}_2 < 80 \text{ GPa}, \tag{19}$$

$$\bar{E}_3 < 80 \text{ GPa}. \tag{20}$$

Correspondingly, the intact set is comprised of the remaining samples $\mathbb{D}_{\text{int}} = \mathbb{D} - \mathbb{D}_{\text{def}}$, which are used to train the GAN.

After training, the discriminator is evaluated on both the intact set $\mathbb{D}_{\text{int}}$ (black) and the defective set $\mathbb{D}_{\text{def}}$ (red), depicted in Fig. 6(a). Both sets yield separate clusters. Similar clusters can be observed for the Young's moduli in Fig. 6(b). While in Fig. 6 the effective Young's modulus $\bar{E}_1$ is plotted against the absolute frequency of occurrence, in Fig. 7 the effective properties $\bar{E}_1, \bar{E}_2, \bar{E}_3$ are plotted against the discriminator output.

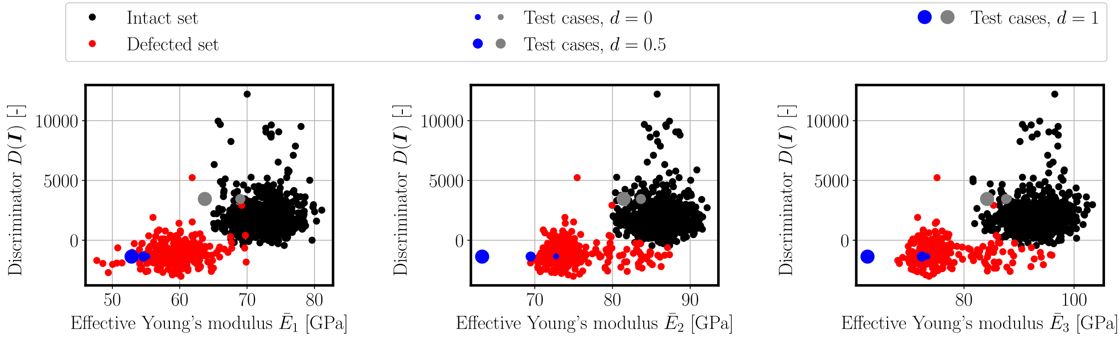
The clusters can be separated by selecting a classification boundary defined for the discriminator output. In this work, the sum of the false positive rate and false negative rate of identifying defective structures as intact is minimized. The classification boundary is thereby determined to be $D^c = 864$. Thus, a simple classification rule can be formulated:

- intact if $D(\boldsymbol{I}) \geq D^c$,
- defective if $D(\boldsymbol{I}) < D^c$.

Based on the data $\mathbb{D}$, this rule leads to

- a false positive rate of 1.87%,
- a false negative rate of 9.13%.

The critical rate is the false positive rate, as incorrectly classifying a defective sample as intact is much more severe than mislabeling an intact sample as flawed. A false positive rate of 1.87% is deemed appropriate. A more in-depth quantification of the anomaly detection performance is given by a receiver operating characteristic (ROC) curve in Appendix A.



(a) x_1 direction (b) x_2 direction (c) x_3 direction

Fig. 7. Discriminator evaluation plotted against the effective Young's moduli in x_1, x_2, and x_3 direction. In analogy to Fig. 6, blue dots refer to manipulated structures from the defected set $\mathbb{D}_{\mathrm{def}}$, while gray dots refer to intact set $\mathbb{D}_{\mathrm{int}}$.

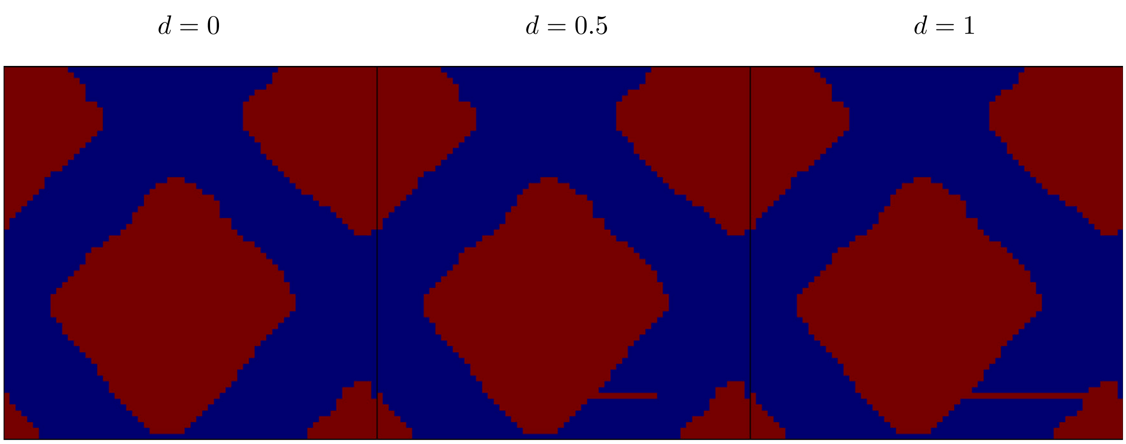


Fig. 8. Test case from the defective set $\mathbb{D}_{\mathrm{def}}$ with a disconnected beam. The disconnection is scaled by the parameter d.

Furthermore, note that our employed GAN architecture is a lightweight variant specifically designed for training on consumer graphic cards. (Karras et al., 2019, 2020, 2021; Hestness et al., 2017). The extensions presented in Karras et al. (2019, 2020) and Karras et al. (2021) were introduced for two-dimensional images and, while adding computational complexity, led to better generative quality. In order to achieve such a powerful generator, the discriminator must be equally good at spotting deviations from the desired norm. Therefore, it would further enhance the discriminative process and, thereby, the anomaly detection capabilities. Nevertheless, the extensions can also lead to overfitting. This can be countered by (a) using more training data, which could become problematic in industrial settings, or by (b) reducing the parameters of the respective layers. In addition, a thorough hyperparameter search is estimated to improve the results further, since in practice one would select the best performing model for production. However, note that a hyperparameter search may be prohibitively computationally expensive in a production setting, since a GAN needs to be trained for each individual part one seeks to monitor.

3.2. Limitations and pitfalls

While the results with real-world experimental data from Section 3.1 demonstrate that the proposed methodology is suitable for application in the context of AM, we now discuss its limitations and caveats. Although the GAN training data has been composed by mechanical considerations, the GAN only learns to detect geometric deviations. Using artificial microstructures, we show that the GAN discriminator is always sensitive to global geometric changes, no matter to what extend they affect the effective mechanical properties (Section 3.2.2). However, local changes are not reliably recognized, even if they have a drastic influence on effective properties see Section 3.2.1. Note that the examples considered in this section are purely artificial and not expected to occur in practical AM applications. Nevertheless, the behavior demonstrated in these examples must be taken into account in practical applications.

3.2.1. Lattice structure with artificial defects

To this end, we consider two samples: one from the defective set $\mathbb{D}_{\mathrm{def}}$ with $D(\boldsymbol{I}) = -1\,348$ and one from the intact set $\mathbb{D}_{\mathrm{int}}$ with $D(\boldsymbol{I}) = 3\,452$. In order to challenge the discriminator, we introduce highly localized artificial disconnections into these structures, see Fig. 8. The parameter d controls the degree of disconnection, where $d = 0$ represents no disconnection and $d = 1$ a full disconnection. The disconnection reduces the load-bearing capacity of the lattice substructure significantly, as seen in the reduction in the Young's modulus in Fig. 6(b) (marked in blue and gray).

However, as can be seen from Fig. 6(a) and Fig. 7, the discriminator output is not sensitive to the level of disconnection d. The corresponding discriminator outputs result in

- $D(\boldsymbol{I}) = -1\,348$ ($\mathbb{D}_{\mathrm{def}}$) and $D(\boldsymbol{I}) = -3\,452$ ($\mathbb{D}_{\mathrm{int}}$) for $d = 0$,
- $D(\boldsymbol{I}) = -1\,358$ ($\mathbb{D}_{\mathrm{def}}$) and $D(\boldsymbol{I}) = -3\,402$ ($\mathbb{D}_{\mathrm{int}}$) for $d = 0.5$,
- $D(\boldsymbol{I}) = -1\,366$ ($\mathbb{D}_{\mathrm{def}}$) and $D(\boldsymbol{I}) = -3\,446$ ($\mathbb{D}_{\mathrm{int}}$) for $d = 1$.

which is almost negligible over the full range of discriminator evaluations from $-3\,000$ to $12\,000$. The same behavior is observed for both of the manipulated microstructures.

These results demonstrate that the GAN learns global features instead of local changes. In order to reliably classify local geometric

Table 3
Geometric characteristics of the artificial microstructures test case of the training data set and two different test sets. The radius R is modified in the first row, and in the second, the volume fraction f is modified. The structures are manipulated by a smaller and a larger modification in the two corresponding quantities.

	Training data set $\mathbb{D}$ and generated set $\mathbb{D}_{gen}$	Small manipulation	Large manipulation
Structures with modified radius R [mm]	4	5	6
Structures with modified volume fraction f [-]	0.2	0.3	0.4

defects, a spatial tensor must be evaluated instead of a scalar quantity. In the literature, different CNN architectures have been developed for this purpose, see, *e.g.*, Strohmann et al. (2021) and Liu et al. (2019) for CNNs in the context of crack detection.

Note that the presented limitation has limited practical implications in the context of AM. In fact, it is very challenging to produce the defects depicted in Fig. 8 using an AM process. Realistic defects are illustrated in Appendix B, and 98.23% of such defects were correctly identified as discussed in Section 3.1.

3.2.2. Artificial microstructure

In this section, we consider an artificial microstructure with spherical voids, see Fig. 4(b) to showcase the opposite effect observed in Section 3.2.1. At constant volume fraction, changes to the void geometry, like the radii, have little influence on the homogenized Young's moduli. Yet, the discriminator still detects these alterations. The data set under investigation consists of samples with volume fraction $f = 0.2$ and radii $R = 4$ mm. It is comprised of 1 000 samples of size $32 \times 32 \times 32$. The artificial microstructures are generated by the *Random sequential adsorption method* algorithm described in Henkes et al. (2021). With this data set, a new instance of the GAN discriminator is trained.

Fig. 9 shows the distribution of the discriminator output $D(\boldsymbol{I})$ (Fig. 9(a)) and the effective Young's modulus $\bar{E}_3$ (Fig. 9(b)) of the structures in the training data set. In order to test the ability of the discriminator to detect geometric outliers, which might cause deviations in the effective properties, the discriminator outputs and computed effective properties of manipulated microstructures are included in the same Fig. 9. The manipulated structures are referred to as test cases and consist of perturbations of the spherical inclusions. First, the radii are increased to $R = 5, 6$ while retaining the same volume fraction $f = 0.2$ (shown in red). Next, the volume fraction is increased to $f = 0.3, 0.4$ while maintaining the same radius $R = 4$ (shown in blue). The specifications of the training data set and test cases are summarized in Table 3.

The test cases without perturbation, *i.e.*, $R = 4$ and $f = 0.2$, lie within the distribution of the training data set for both the discriminator (Fig. 9(a)) and the effective properties (Fig. 9(b)). As the level of the perturbation increases, the discriminator correctly identifies the modified geometries. These have a significantly lower discriminator output (below 400, whereas the entire training data set lies over 800). This is similarly observable in the effective properties when the volume fraction f increases, leading to decreasing effective Young's moduli.

However, note that not every geometrical change necessarily affects the homogenized Young's moduli to a critical extent. This can be seen from the red bars in Fig. 9(b) and the red dots in Fig. 10: At constant volume fraction, a change in spherical radii does not significantly influence the effective Young's moduli. Thus, the discriminator detects geometrical changes independently of whether they are correlated with the effective properties or not. Note, however, that other effective properties, such as those connected to fatigue, would be affected by the size of the spherical inclusions, as known from fundamental studies relating pore size to number of fatigue cycles (Sundström, 2010; Anderson, 2017). While it would be equally possible to pre-select microstructures for GAN training by additional criteria, *e.g.*, related to fatigue, this is considered out of scope of the present manuscript.

The implications of this observation depend on the intended use of the discriminator:

(i) **Reliability indicator**: If the discriminator is used as a reliability indicator upstream to a CNN that has been trained on the same as-built geometry dataset as the GAN, this feature is absolutely desired. For larger radii, a CNN would necessarily extrapolate effective properties, and little to nothing is known about the accuracy in extrapolation.
(ii) **Property monitoring**: On the other hand, the discriminator fails to classify desired and undesired structures by means of target effective properties. This is because the discriminator is physics-agnostic and not aware of the effective properties. Viewing this from a positive standpoint, the discriminator is able to detect defects unseen in the effective properties — which could potentially be detrimental with regard to other unobserved properties.

While the test case with spherical inclusions provides insights into the capabilities and limitations of the proposed workflow, it must be emphasized that it is purely of academic interest. A real-world AM test case with actual defects, as considered in Section 3.1, results in a GAN yielding satisfactory results, both as reliability indicator as well as for property monitoring.

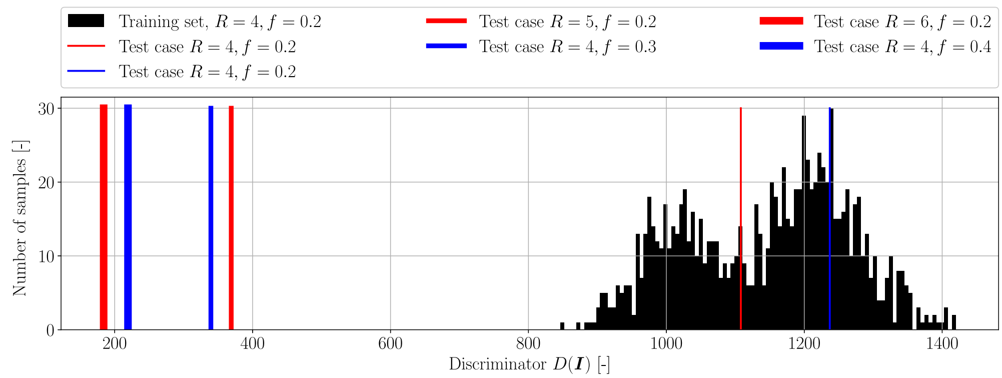
4. Conclusion

The process inherent stochastics of AM may result in significant uncertainties in geometric accuracy, which in turn may have a considerable influence on effective properties. This is especially problematic in the design of complex structures, for which AM is especially suited. Hence, there is a huge need for quality control and part certification. In this work, we propose a GAN-based approach to outlier detection and property monitoring for AM.

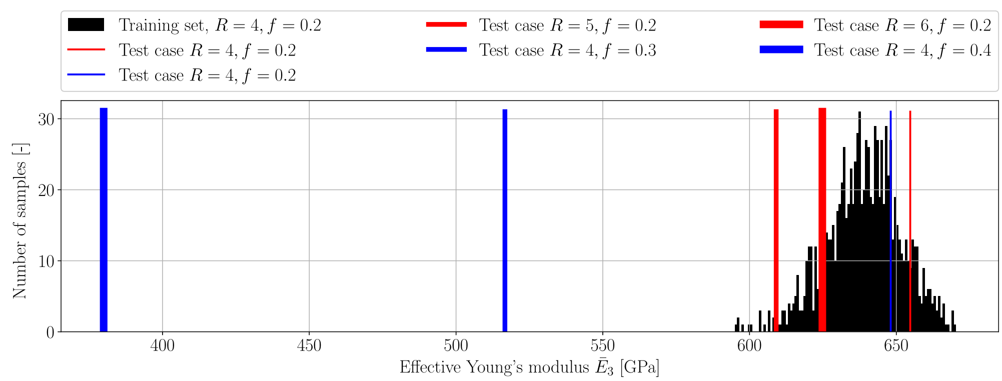
To this end, the discriminator of a Wasserstein GAN is utilized to detect as-built outliers in both synthetic and experimental data sets. If used in conjunction with CNN homogenization, the GAN discriminator comes at no extra computational cost and can be regarded as reliability indicator for CNN predictions. In this manuscript, the following steps of this workflow are considered: (i) effective mechanical properties for experimentally obtained images of as-built geometries were obtained with the FCM, (ii) admissible as-built geometries were augmented using a GAN, (iii) the GAN discriminator provides an anomaly score for outlier detection.

For realistic AM structures and defects considered in this manuscript, the presented GAN shows excellent performance. This is especially satisfactory in view of the comparatively lightweight GAN architecture employed and the relatively small experimental database used for training. Therefore, the presented methodology can provide valuable insights for automated outlier detection and property monitoring in real-world AM processes.

Besides this success, also limitations and pitfalls were thoroughly discussed. For this purpose, artificial test cases were designed that are

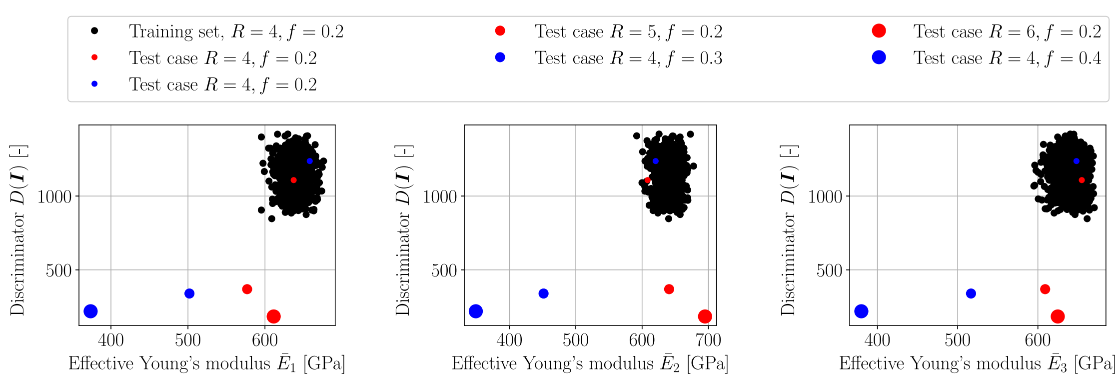


(a) Discriminator evaluation



(b) Effective Young's moduli

Fig. 9. Comparison of the distribution of discriminator evaluations (a) and effective Young's moduli in the x_3 direction (b). The testing set includes samples belonging to the distribution of the training set and perturbed samples. The perturbation is introduced by varying the radius (red) and volume fraction (blue) correspondingly. The magnitude of the perturbation is indicated by the thickness of the bars.



(a) x_1 direction (b) x_2 direction (c) x_3 direction

Fig. 10. Discriminator evaluation plotted against the effective Young's moduli in x_1, x_2, and x_3 direction. The perturbation is introduced by varying the radius and volume fraction correspondingly. The magnitude of the perturbation is indicated by the thickness of the markers. The discriminator detects the geometric outliers very well. Note that the geometric deviation only yields a significant change in property for the perturbations in the volume fraction.

especially difficult to handle for the GAN considered. It became apparent that although the GAN training data was selected upon mechanical properties, the discriminator detects solely geometric features: while highly localized defects (thin cuts) with a large impact on mechanical properties were not detected, global deviations with little effect, in turn, were classified as defective. Note that the considered artificial defects are not expected to occur within real-world AM processes. Nevertheless, these findings should be carefully taken into account when implementing a GAN discriminator into an actual production setting. Moreover, these results also motivate future work, *i.e.*, to develop methodologies that account for the monitoring and assessment of properties during production in order to further increase the robustness of outlier detection.

Data and code availability

- The basis for GAN code has been published in Henkes and Wessels (2022b).

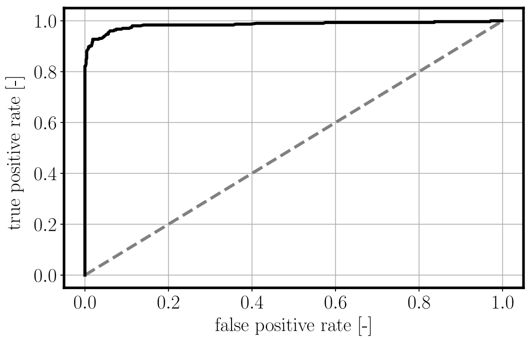


Fig. 11. ROC curve of the discriminator for anomalous lattice structures yielding an AUC of 0.983.

- A publicly available FCM code can be found in Zander et al. (2014).
- The artificial microstructures and the CT scan are made available at Henkes et al. (2024).

CRediT authorship contribution statement

Alexander Henkes: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Leon Herrmann:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Henning Wessels:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Stefan Kollmannsberger:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

Alexander Henkes was supported by an ETH Zurich Postdoctoral Fellowship. Henning Wessels gratefully acknowledges funding of the German Research Foundation (DFG), Germany for the individual research grant 512730472 entitled “Data-driven simulation of microstructure in powder bed fusion processes”. Furthermore, Stefan Kollmannsberger acknowledges the support of Deutsche Forschungsgemeinschaft (DFG, Germany) through the project 414265976 TRR 277 C-01.

Appendix A. Receiver operating characteristic curve of lattice anomaly detection

In order to quantify the anomaly classification capabilities, the receiver operating characteristic (ROC) curve is provided in Fig. 11, yielding an area under the curve (AUC) of 0.983. The ROC curve is constructed by computing the true and false positive rates with different thresholds ranging between the smallest and largest obtained Wasserstein distance.

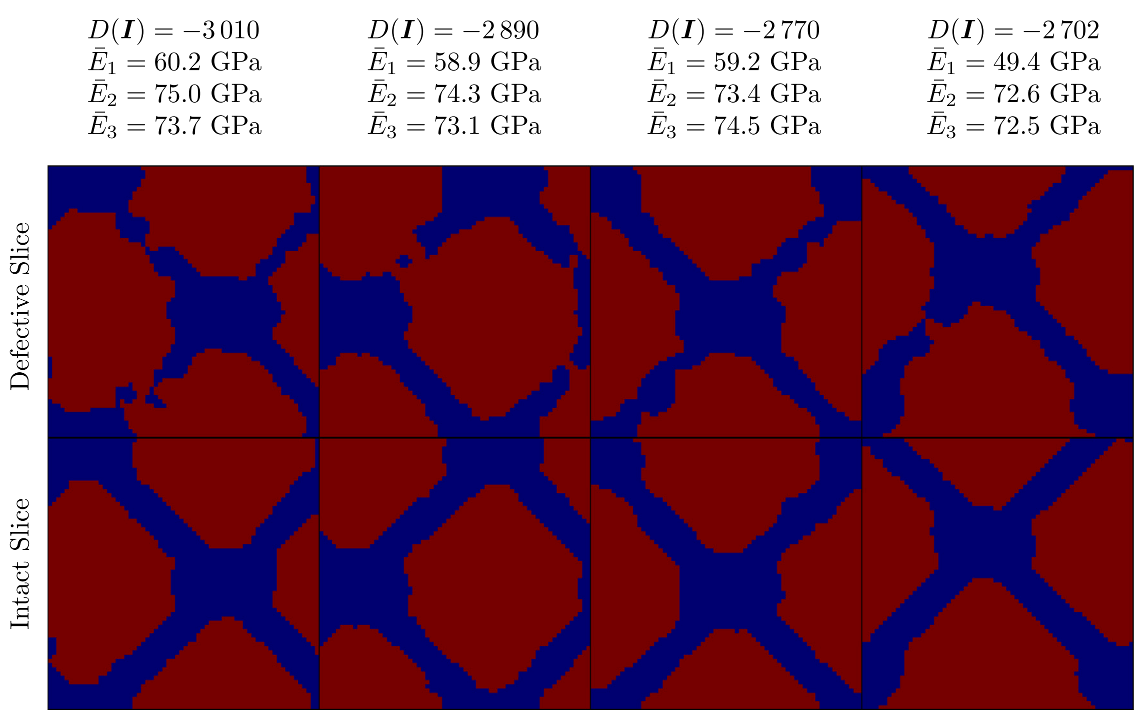


Fig. 12. Slices of realistic defects present in the lattice structure from Fig. 5. These are part of the defective set $\mathbb{D}_{\text{def}}$. Each column shows two slices of the same specimen. The top row showcases defects, such as partial disconnections, while the lower row showcases typical slices also encountered in intact structures, *i.e.*, in the intact set $\mathbb{D}_{\text{int}}$.

Appendix B. Realistic defects in lattice structures

In contrast to the artificial disconnections presented in Fig. 8, Fig. 12 shows representative examples of real defects encountered in the investigated lattice structure (Fig. 5). These defects lead to reduced effective properties and have been correctly identified by the discriminator. Defects that affect the effective properties are typically partial disconnections or thinned-out connections.

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